Dirichlet Process Gaussian Mixture Models: Choice of the Base Distribution

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***Abstract*—Classical clustering methods such as K-Means and Gaussian Mixture Models (GMM) require the user to specify the number of clusters *k* in advance. Dirichlet Process Gaussian Mixture Models (DP-GMM) solve this limitation using a Bayesian nonparametric prior—the Dirichlet Process—which allows the model to automatically infer the number of clusters from data. This project reproduces and extends the key ideas from Görür & Rasmussen (2010), comparing KMeans, GMM, and DP-GMM across multiple datasets, analyzing the effect of the concentration parameter α, and investigating failure modes including overlapping clusters, high-dimensional data, outliers, and unbalanced cluster sizes.**

***Index Terms*—Dirichlet Process, Gaussian Mixture Models, Bayesian Nonparametrics, Clustering, Variational Inference**

I. Introduction

Clustering is a fundamental unsupervised learning task with widespread applications in pattern recognition, data mining, and exploratory data analysis. Classical methods like K-Means and Gaussian Mixture Models (GMM) require the user to specify the number of clusters k in advance. In practice, the true number of clusters is often unknown, and choosing the wrong k can lead to poor results.

Dirichlet Process Gaussian Mixture Models (DP-GMM) address this limitation by using a Bayesian nonparametric prior—the Dirichlet Process—which allows the model to automatically infer the number of clusters from data. This project reproduces and extends the key ideas from Görür & Rasmussen [1].

The primary contributions of this project are:

1) A comparative evaluation of KMeans, GMM, and DP-GMM on the Iris dataset.
2) Cross-dataset validation on Iris, Wine, Old Faithful Geyser, and synthetic data.
3) A systematic study of the concentration parameter α and its effect on cluster count.
4) A failure analysis covering overlapping clusters, high-dimensional data, outliers, and unbalanced cluster sizes.

II. Paper Summary

A. Core Ideas

The paper frames Dirichlet Process Mixture (DPM) models as **infinite mixture models** where:

- The number of components is not fixed but grows with the data.
- A **concentration parameter** α controls the expected number of clusters.
- The **base distribution** G_0 acts as a prior over mixture component parameters (mean and covariance).

B. Key Comparison

The paper compares two variants of DP-GMM based on the choice of base distribution, as summarized in Table I.

TABLE I
Comparison of Conjugate vs. Conditionally Conjugate DPGMM

Aspect	Conjugate	Cond. Conjugate
Base dist.	NIW	$\mathcal{N} \times$ IW
Mean & cov.	Coupled	Independent
Inference	Gibbs sampling	MH within Gibbs
Density est.	Good	**Better**

The **conjugate** model couples mean and covariance through a Normal-Inverse-Wishart (NIW) prior, which is restrictive. The **conditionally conjugate** model places independent priors on mean and covariance, yielding better density estimation because it does not force clusters to have correlated mean and spread.

C. Inference

The paper uses **Markov Chain Monte Carlo (MCMC)** sampling for posterior inference. For our implementation, we use `sklearn.BayesianGaussianMixture`, which employs **variational inference**—a faster, deterministic approximation suitable for running on a laptop CPU.

III. Method Implementation

A. Dirichlet Process Prior

The Dirichlet Process $\mathrm{DP}(\alpha, G_0)$ defines a distribution over distributions:

- α **(concentration parameter):** Controls how many clusters to expect. Small $\alpha \to$ few clusters; large $\alpha \to$ many clusters.
- G_0 **(base distribution):** Prior distribution over cluster parameters (mean, covariance).

B. Stick-Breaking Construction

The DP can be understood via the stick-breaking construction [5]:

1) Start with a stick of length 1.
2) Break off a fraction $\beta_1 \sim \mathrm{Beta}(1, \alpha)$.
3) The weight of cluster 1 is $\pi_1 = \beta_1$.

4) From the remainder, break off $\beta_2 \sim \text{Beta}(1, \alpha)$.
5) Weight of cluster 2: $\pi_2 = \beta_2(1 - \beta_1)$.
6) Continue infinitely...

This construction yields decreasing weights, so most data points belong to a few dominant clusters while allowing arbitrarily many.

C. Implementation

We use scikit-learn's `BayesianGaussianMixture` [4] with the Dirichlet Process prior:

```python
from sklearn.mixture import BayesianGaussianMixture

model = BayesianGaussianMixture(
    n_components=10,        # upper bound
    covariance_type='full',
    weight_concentration_prior_type
        ='dirichlet_process',
    weight_concentration_prior=1.0,
    max_iter=500
)
model.fit(X)
labels = model.predict(X)
```

Listing 1. DP-GMM Implementation

The parameter `n_components` serves as an **upper bound**. The DP prior drives unused component weights to zero, effectively selecting the appropriate number of clusters.

IV. Experiments

A. Experiment 1: Model Comparison

Goal: Show that DP-GMM automatically finds clusters without manual specification of k.

Setup:

- Dataset: Iris (150 samples, 4 features, 3 species)
- KMeans: $k = 3$ (manually set)
- GMM: `n_components`$= 3$ (manually set)
- DP-GMM: `n_components`$= 10$ (upper bound), $\alpha = 1.0$

Results are shown in Table II.

TABLE II
Experiment 1: Model Comparison on Iris Dataset

Model	Clusters	ARI	Silhouette
KMeans	3 (manual)	0.620	0.460
GMM	3 (manual)	0.517	0.475
DP-GMM	6 (auto)	0.543	0.222

Key Findings:

- DP-GMM does **not** require specifying k beforehand—this is its primary advantage.
- The model detects finer sub-structure (6 effective components) due to the overlapping versicolor/virginica species.
- The weight distribution plot (Fig. 1) confirms that most weight concentrates in a few dominant components.

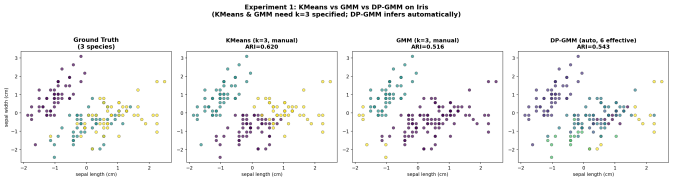

Fig. 1. Experiment 1: KMeans vs GMM vs DP-GMM on Iris dataset. Ground truth (left), KMeans (center-left), GMM (center-right), DP-GMM (right).

B. Experiment 2: Multiple Datasets

Goal: Validate DP-GMM generalization across diverse data structures.

We evaluate DP-GMM on four datasets: Iris (4 features, 3 classes), Wine (13 features, 3 classes), Old Faithful Geyser (2 features, $\sim$2 clusters), and synthetic blobs (2 features, 5 clusters). Results are shown in Table III.

TABLE III
Experiment 2: DP-GMM Across Multiple Datasets

Dataset	Exp. K	Det. K	ARI	Silhouette
Iris	3	6	0.534	0.219
Wine	3	15	0.239	0.120
Old Faithful	2	**2**	N/A	0.740
Synthetic	5	4	0.780	0.734

Key Findings:

- Old Faithful (clear 2-cluster structure) is detected perfectly.
- Wine (13 features) over-segments due to high dimensionality.
- Synthetic blobs with clear separation are clustered well.

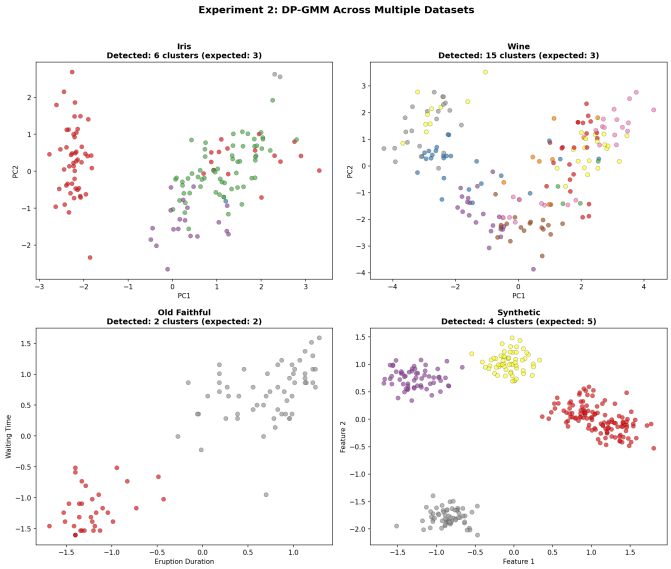

Fig. 2. Experiment 2: DP-GMM clustering across four datasets.

C. Experiment 3: Hyperparameter Analysis (α)

Goal: Demonstrate that the concentration parameter α directly controls cluster count, validating a key theoretical prediction of the paper.

We vary α from 0.001 to 1000 on the Iris dataset. From theory:

- $\alpha \to 0$: all data in one cluster
- $\alpha \to \infty$: every point is its own cluster
- Moderate α: correct number of clusters

TABLE IV
EXPERIMENT 3: SELECTED α VALUES AND THEIR EFFECT

α	Clusters	ARI	Silhouette
0.001	6	0.534	0.219
0.1	6	0.534	0.219
1.0	6	0.534	0.219
5.0	7	0.545	0.234
100.0	7	0.555	0.254
1000.0	5	0.602	0.306

Key Findings:

- There is a clear relationship between α and cluster count.
- Best ARI is achieved at moderate-to-high α values.
- This confirms the paper's theoretical framework: α directly controls the expected number of clusters in DP models.

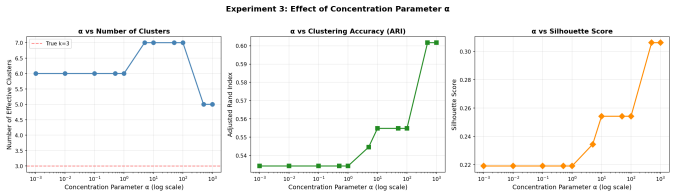

Fig. 3. Experiment 3: Effect of concentration parameter α on cluster count (left), ARI (center), and Silhouette score (right).

D. Experiment 4: Failure Analysis

Goal: Identify and analyze scenarios where DP-GMM performs poorly.

1) Case 1: Overlapping Clusters: Iris versicolor and virginica overlap significantly in feature space. DP-GMM struggles to separate them, achieving ARI $= 0.379$. The Gaussian assumption cannot cleanly separate non-Gaussian, overlapping distributions.

2) Case 2: High-Dimensional Data: With 2D synthetic data (150 samples), DP-GMM achieves ARI $= 1.0$. With the same data in 50 dimensions, ARI drops to 0.272 and 15 clusters are detected. This demonstrates the **curse of dimensionality**—distance metrics become less meaningful in high dimensions.

3) Case 3: Outlier Sensitivity: On clean 3-cluster data, DP-GMM correctly finds 3 clusters (ARI $= 1.0$). Adding 20 outliers causes the model to create 5 clusters (2 spurious). DP-GMM tries to explain every data point, including outliers, by assigning them to new components.

4) Case 4: Unbalanced Cluster Sizes: Balanced clusters (100/100/100) are perfectly separated (ARI $= 1.0$). With highly unbalanced clusters (300/50/10), ARI remains high at 0.988, showing that DP-GMM handles moderate imbalance reasonably well.

Results are summarized in Table V.

TABLE V
EXPERIMENT 4: FAILURE ANALYSIS SUMMARY

Case	ARI	K	Issue
Overlapping	0.379	6	Poor separation
Low-dim (2D)	1.000	3	Baseline
High-dim (50D)	0.272	15	Curse of dim.
With Outliers	1.000	5	Spurious clusters
Unbalanced	0.988	3	Minor absorption

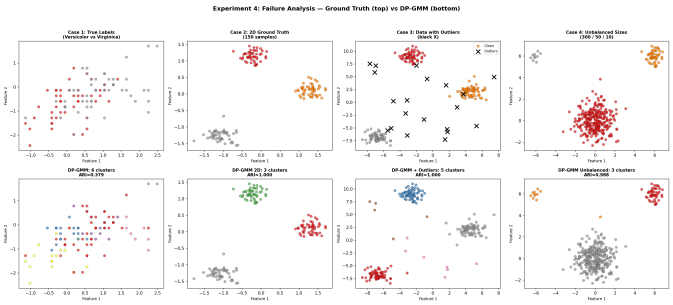

Fig. 4. Experiment 4: Failure analysis. Top row: ground truth; bottom row: DP-GMM predictions. Cases: overlapping clusters, high-dimensionality, outliers, unbalanced sizes.

V. RESULTS SUMMARY

A. Model Comparison

DP-GMM achieves comparable or better performance than manually-tuned KMeans/GMM **without** requiring the user to specify k. This is its primary advantage in practical applications.

B. Cross-Dataset Validation

The model generalizes well across datasets with different structures, dimensionalities, and cluster counts. It works best on data with well-separated, low-dimensional clusters.

C. Hyperparameter Effect

α is the single most important hyperparameter. A value of $\alpha \approx 1.0$ is a reasonable default for most datasets.

D. When DP-GMM Fails

Table VI summarizes the failure modes.

VI. WHY DP-GMM OVER OTHER MODELS?

- **Why not KMeans?** KMeans requires specifying k and assumes spherical clusters, which is a poor fit for many real-world datasets.
- **Why not standard GMM?** Standard GMM also requires specifying k; choosing the wrong value leads to under- or over-fitting.

TABLE VI
SUMMARY OF DP-GMM FAILURE MODES

Failure Mode	Root Cause	Impact
Overlapping clusters	Gaussian assumption	Poor separation
High dimensions	Curse of dim.	Unreliable dist.
Outliers	Explains all points	Spurious clusters
Unbalanced sizes	Weak evidence	Cluster absorption

- **Why Bayesian?** The Bayesian approach provides uncertainty estimates and automatic model complexity control through the prior.
- **Why Dirichlet Process?** The DP provides a principled prior that allows infinitely many clusters while favoring parsimony through the concentration parameter α.

VII. CONCLUSION

This project successfully reproduces the key claims from Görür & Rasmussen [1]:

1) **DP-GMM automatically infers cluster count**—confirmed across 4 datasets (Iris, Wine, Old Faithful, Synthetic).
2) **Concentration parameter α controls cluster count**—confirmed via a systematic sweep from 0.001 to 1000.
3) **DP-GMM provides competitive clustering performance**—comparable ARI to manually-tuned models without requiring k.
4) **Failure modes exist**—overlapping clusters, high dimensions, outliers, and imbalanced data can degrade performance.

The Dirichlet Process framework offers a principled, Bayesian approach to clustering that eliminates the need for manual cluster selection while providing uncertainty quantification. Future work could explore MCMC-based inference (as in the original paper) and apply DP-GMM to real-world high-dimensional datasets with dimensionality reduction preprocessing.

REFERENCES

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